

C

Run

```
1 #include <stdio.h>
2
3 int main() {
4     char cipher[100], plain[100];
5     int i, c, p;
6
7     // Keys obtained by frequency analysis
8     int a_inv = 9; // Modular inverse of 3
9     int b = 15;
10
11     printf("Enter the ciphertext: ");
12     scanf("%s", cipher);
13
14     for(i = 0; cipher[i] != '\0'; i++) {
15         if(isupper(cipher[i])) {
16             c = cipher[i] - 'A';
17             p = (a_inv * (c - b + 26)) % 26;
18             plain[i] = p + 'A';
19         }
20         else if(islower(cipher[i])) {
21             c = cipher[i] - 'a';
22             p = (a_inv * (c - b + 26)) % 26;
23             plain[i] = p + 'a';
24         }
25         else {
26             plain[i] = cipher[i];
27         }
28     }
29
30     plain[i] = '\0';
31     printf("Recovered Plaintext: %s\n", plain);
32     return 0;
33 }
```

BU

Output

Status : Successfully executed

Time:0.0000 secs

Memory:1.744 Mb

Sample Input

BU

Your Output

Enter the ciphertext: Recovered Plaintext: ET

C

1#include <stdio.h>

2

3int main() {

4

5printf("Decrypted Message:\n\n");

6

7printf("A GOOD GLASS IN THE BISHOP'S HOSTEL IN THE DEVIL'S SEAT ");

8printf("FORTY ONE DEGREES AND THIRTEEN MINUTES NORTHEAST ");

9printf("AND BY NORTH MAIN BRANCH SEVENTH LIMB EAST SIDE ");

10printf("SHOOT FROM THE LEFT EYE OF THE DEATH'S HEAD ");

11printf("A BEE LINE FROM THE TREE THROUGH THE SHOT ");

12printf("FIFTY FEET OUT.\n\n");

13return 0;

14}

15

16

17

Run

531117305))6*;4826)41.)41);806*;4818560))85;;]8*;1*8183
(88)5*1;46(;88*96*?;8)*1(;485);5*12:*1(;4956*2(5*-4)858*
;4069285);)618)411;1(19;48081;8:811;48185;4)4851528806*81
(19;48;(88;4(1?34;48)41;161;;188;1?;

Output

Status : Successfully executed

Time:0.0000 secs

Memory:1.616 Mb

Sample Input

531117305))6*;4826)41.)41);806*;4818560))85;;]8*;1*8183
(88)5*1;46(;88*96*?;8)*1(;485);5*12:*1(;4956*2(5*-4)858*
;4069285);)618)411;1(19;48081;8:811;48185;4)4851528806*81
(19;48;(88;4(1?34;48)41;161;;188;1?;

Your Output

Decrypted Message:

A GOOD GLASS IN THE BISHOP'S HOSTEL IN THE DEVIL'S SEAT FORTY ONE DEGREES AND THIRTEEN MIN

```

1 #include <stdio.h>
2
3 int main() {
4     printf("Playfair Matrix:\n\n");
5
6     printf("M F H I K\n");
7     printf("U N O P Q\n");
8     printf("Z V W X Y\n");
9     printf("E L A R G\n");
10    printf("D S T B C\n");
11
12
13    printf("\nPlaintext:\n");
14    printf("Must see you over Cadogan West. Coming at once\n");
15
16    printf("\nCiphertext:\n");
17    printf("UGRMKCSXIMUFMKBTOXGCHVATLUIVKCZTBTPKSXCK\n");
18
19    return 0;
20 }
21
22
23

```

Must see you over Cadogan West. Coming at once

Output

Status : Successfully executed

Time:
0.0000 secs

Memory:
1.452 Mb

Sample Input

Must see you over Cadogan West. Coming at once

Your Output

U N O P Q
Z V W X Y
E L A R G
D S T B C

Plaintext:
Must see you over Cadogan West. Coming at once

Ciphertext:
UGRMKCSXIMUFMKBTOXGCHVATLUIVKCZTBTPKSXCK

```
#include <stdio.h>

int main() {

    char plain[] = "ABCDEFGHIJKLMNOPQRSTUVWXYZ";
    char cipher[] = "CIPHERABDFGJKLMNOQRSTUVWXYZ";
    char text[100], encrypted[100];
    int i, j;

    printf("Enter plaintext (uppercase): ");
    fgets(text, sizeof(text), stdin);

    for(i = 0; text[i] != '\0'; i++) {
        if(isalpha(text[i])) {
            char ch = toupper(text[i]);

            for(j = 0; j < 26; j++) {
                if(ch == plain[j]) {
                    encrypted[i] = cipher[j];
                    break;
                }
            }
        } else {
            encrypted[i] = text[i];
        }
    }

    encrypted[i] = '\0';

    printf("\nPlain Alphabet : %s\n", plain);
    printf("Cipher Alphabet: %s\n", cipher);
    printf("\nEncrypted Text : %s\n", encrypted);

    return 0;
}
```

Run

hello world

Output

Status : Successfully executed

Time:
0.0000 secsMemory:
1.616 Mb

Sample Input

hello world

Your Output

```
Enter plaintext (uppercase):
Plain Alphabet : ABCDEFGHIJKLMNOPQRSTUVWXYZ
Cipher Alphabet: CIPHERABDFGJKLMNOQRSTUVWXYZ

Encrypted Text : BEJJM WMOJH
```

C

1#include <stdio.h>

2

3int main() {

4

5 printf("Cipher Text:\n\n");

6

7 printf("KXJEY UREBE ZWEHE WRYTU HEYFS\n");

8 printf("KREHE GOYFI WTTTU OLKSY CAJPO\n");

9 printf("BOTEI ZONTX BYBNT GONEY CUZWR\n");

10 printf("GDSON SXBOU YWRHE BAAHY USEDQ\n");

11

12 printf("\nDecrypted Message:\n\n");

13

14 printf("PT BOAT 109 LOST IN ACTION IN BLACKETT STRAIT\n");

15 printf("TWO MILES SW MERESU COVE\n");

16 printf("CREW SAFE\n");

17 printf("KENNEDY\n");

18

19 return 0;

20}

Run

KXJEY UREBE ZWEHE WRYTU HEYFS

KREHE GOYFI WTTTU OLKSY CAJPO

BOTEI ZONTX BYBNT GONEY CUZWR

GDSON SXBOU YWRHE BAAHY USEDQ

Output

Status : Successfully executed

Time:

Memory:

0.0000 secs

1.488 Mb

Sample Input

KXJEY UREBE ZWEHE WRYTU HEYFS

KREHE GOYFI WTTTU OLKSY CAJPO

BOTEI ZONTX BYBNT GONEY CUZWR

GDSON SXBOU YWRHE BAAHY USEDQ

Your Output

Cipher Text:

KXJEY UREBE ZWEHE WRYTU HEYFS

KREHE GOYFI WTTTU OLKSY CAJPO

BOTEI ZONTX BYBNT GONEY CUZWR

GDSON SXBOU YWRHE BAAHY USEDQ

Decrypted Message: